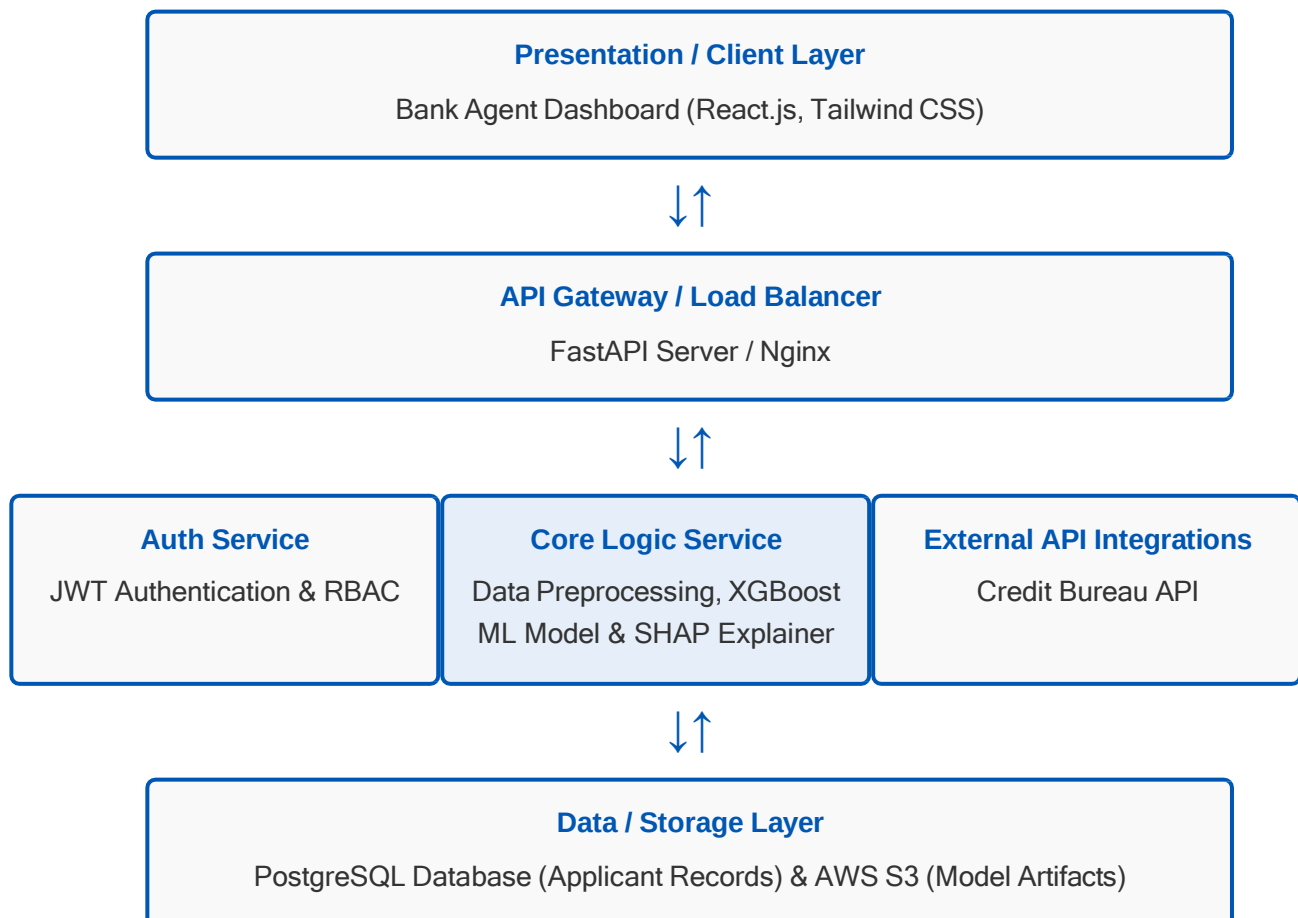


Date	15 March 2026
Team ID	xxxxxx
Project Name	Credit Card Approval Prediction
Maximum Marks	5 Marks

Solution Architecture Diagram:

Provide a high-level visual representation of your solution's architecture. Use the structural layout below as a template to map out your Presentation Layer, Application Layer, and Data Layer.



Component Description Table

Component Name	Description / Role in Architecture	Technologies Used
Presentation Layer	Provides the user interface for bank agents to securely input applicant data and view real-time ML approval predictions alongside SHAP explainability reports.	React.js, Tailwind CSS
API Gateway / Server	Routes incoming dashboard requests, manages load, and serves the machine learning model prediction endpoints securely.	FastAPI, Nginx
Core Logic Service (ML)	Handles raw financial data preprocessing, executes the XGBoost classification model to predict approval probabilities, and calculates SHAP values.	Python, Scikit-Learn, XGBoost, SHAP
Database / Data Storage	Stores historical applicant datasets, user credentials, prediction logs, and hosts the serialized machine learning model files for deployment.	PostgreSQL, AWS S3